A two-use history advantage under one-bit coarse-grained environment feedback

Xiaojian Zeng
PBC School of Finance, Tsinghua University, Beijing 100190, China
xiaojianzeng@foxmail.com

Abstract

Environment-assisted recovery is usually formulated with every measured Kraus outcome resolved. We instead disclose one bit obtained by pairing four outcomes of a full-rank qubit channel. For two uses of an explicit generalized-amplitude-damping instrument, we prove that every controller whose recovery sees only the current bit has entanglement fidelity at most 456/625. A controller retaining the preceding bit attains 1831/2500, giving a certified advantage of at least 7/2500. Since the two policy classes coincide for one use, two uses are the earliest possible depth at which coarse-record history can help. The upper bound permits arbitrary qubit CPTP recoveries.

Keywords: environment-assisted correction; quantum feedback; coarse-grained measurement; entanglement fidelity; generalized amplitude damping

1 Introduction

Classical information obtained from an environment can guide recovery of a noisy quantum system. In the resolved-record setting, each measured outcome labels a single Kraus operator. Gregoratti and Werner derived optimal discrete-time recovery formulas for this setting [1]. Rexiti, Oskouei, and Mancini subsequently compared current-outcome and history-conditioned discrete feedback and showed that the two have equal performance for their rank-two qubit protocol [2]. Related Bayesian–Markovian distinctions also arise in continuous monitored feedback, but with a different dynamics, observation process, and objective [3].

Finite-bandwidth readout changes the mathematical object presented to the controller. If several orthogonal environment outcomes are assigned the same classical symbol, an observed branch is a multi-Kraus completely positive operation; the unobserved alternatives must be summed incoherently before control. A polar correction for each hidden Kraus operator is then unavailable. We ask the first depth at which retaining the history of such a coarse record can improve entanglement fidelity over all current-bit-only recovery policies.

Our contribution is an exact two-use separation. We give a full-Choi-rank generalized amplitude-damping (GAD) channel and a fixed balanced pairing of its four environment outcomes. A trace-norm data-processing argument bounds each Bloch column attainable after one current-bit correction. Exact factorisation of a two-use current-bit policy, followed by Frobenius Cauchy–Schwarz, yields a global bound over arbitrary qubit CPTP recoveries. An explicit history policy exceeds it. The conclusion concerns the information structure of this fixed finite-dimensional instrument; it is not an experimental robustness or continuous-time statement.

2 Model and policy classes

Let the system be $\mathbb{C}^2$. A fixed minimal Kraus representation $\boldsymbol{K} = (K_0, K_1, K_2, K_3)$ defines

$$\mathcal{T}(\rho) = \sum_{x=0}^{3} K_x \rho K_x^\dagger, \qquad \sum_x K_x^\dagger K_x = I. \tag{1}$$

The representation is physically tied to an orthogonal measurement of a fresh four-dimensional environment. A balanced map $g : \{0,1,2,3\} \to \{0,1\}$ discloses only $b = g(x)$, so the observed instrument is

$$\mathcal{I}_b(\rho) = \sum_{x:g(x)=b} K_x \rho K_x^\dagger. \tag{2}$$

The fine outcome x is never available. Independent fresh environments and the same instrument are used at each step.

After record $b_{1:k}$, a deterministic policy applies a qubit CPTP map $\mathcal{R}_{k,b_{1:k}}$. Its unnormalised branches obey

$$\Gamma_\varnothing = \mathrm{id}, \qquad \Gamma_{b_{1:k}} = \mathcal{R}_{k,b_{1:k}} \circ \mathcal{I}_{b_k} \circ \Gamma_{b_{1:k-1}}, \qquad \Phi_n = \sum_{b_{1:n}} \Gamma_{b_{1:n}}. \tag{3}$$

The current-bit class allows time-dependent $\mathcal{R}_{k,b_k}$, whereas the full-history class allows $\mathcal{R}_{k,b_{1:k}}$. Both use the current bit immediately. Neither class has access to hidden fine outcomes, postselection, or persistent quantum memory. Randomised recovery is already included through CPTP mixing.

Performance is the entanglement fidelity for the maximally mixed qubit,

$$F_e(\Phi) = \langle\Omega|(\mathrm{id} \otimes \Phi)(|\Omega\rangle\langle\Omega|)|\Omega\rangle, \qquad |\Omega\rangle = (|00\rangle + |11\rangle)/\sqrt{2}. \tag{4}$$

Let F_n^M and F_n^B be the maxima in the current-bit and full-history classes and $\Delta_n = F_n^B - F_n^M$. Compactness of the finite products of qubit-CPTP Choi sets ensures that these maxima exist. At $n = 1$ the two policy classes are identical.

3 A full-rank coarse GAD instrument

Set

$$p = \frac{18}{25}, \qquad q = 1 - p = \frac{7}{25}, \qquad \gamma = \frac{3}{10}, \qquad s = \sqrt{1-\gamma} = \sqrt{\frac{7}{10}}, \tag{5}$$

and use the ordered GAD Kraus operators

$$\begin{aligned} K_0 &= \sqrt{p}\ \mathrm{diag}(1, s), & K_1 &= \sqrt{p\gamma}\,|0\rangle\langle 1|, \\ K_2 &= \sqrt{q}\ \mathrm{diag}(s, 1), & K_3 &= \sqrt{q\gamma}\,|1\rangle\langle 0|. \end{aligned} \tag{6}$$

Equation (1) follows by direct diagonal summation. The off-diagonal operators K_1 and K_3 occupy distinct nonzero matrix units. Within the diagonal subspace, the determinant of the two coordinate vectors from K_0 and K_2 is $\sqrt{pq}(1 - s^2) = \sqrt{pq}\,\gamma$, whose squared modulus is

$$pq\gamma^2 = \frac{567}{31250} > 0. \tag{7}$$

Thus the four operators are linearly independent and the channel has Choi rank four.

We disclose the pairing 03|12:

$$\mathcal{I}_0(\rho) = K_0 \rho K_0^\dagger + K_3 \rho K_3^\dagger, \qquad \mathcal{I}_1(\rho) = K_1 \rho K_1^\dagger + K_2 \rho K_2^\dagger. \tag{8}$$

Both observed operations genuinely contain two hidden alternatives.

4 Exact separation

Theorem 1. *For the instrument in Eqs. (5)–(8), every two-use current-bit policy with arbitrary qubit CPTP recoveries satisfies*

$$F_2^M \le \frac{456}{625}. \tag{9}$$

There is a full-history policy with fidelity $1831/2500$. *Consequently* $\Delta_2 \ge 7/2500 > 0$, *and the earliest use count at which a coarse-record history advantage can occur is* $n_* = 2$.

4.1 A feasible full-history policy

Let X be the Pauli bit flip and choose

$$R_{1,0} = I, \quad R_{1,1} = X, \qquad R_{2,b_1b_2} = \begin{cases} I, & b_1 = 0, \\ X, & b_1 = 1. \end{cases} \tag{10}$$

These recoveries are unitary and hence CPTP. Keeping the fine outcomes only as an algebraic device for evaluating the incoherent sums, the entanglement fidelity is

$$F_e = \frac{1}{4} \sum_{x_1,x_2=0}^{3} \left| \mathrm{Tr}\left[R_{2,g(x_1)g(x_2)} K_{x_2} R_{1,g(x_1)} K_{x_1} \right] \right|^2 . \tag{11}$$

Direct multiplication of the sixteen paths leaves the following multiset of nonzero trace amplitudes:

$$p(2-\gamma), \quad 2\sqrt{pq(1-\gamma)}, \quad \gamma\sqrt{pq}, \quad p\gamma, \quad \sqrt{pq}(2-\gamma), \quad 2q\sqrt{1-\gamma}. \tag{12}$$

Substitution into Eq. (11) gives the exact lower bound

$$F_2^B \ge F_{\mathrm{hist}} = 1 - \gamma + \frac{p\gamma^2}{2} = \frac{1831}{2500}. \tag{13}$$

4.2 A global current-bit upper bound

Consider one stage with arbitrary qubit CPTP maps R_0, R_1 and define

$$C = R_0 \circ \mathcal{I}_0 + R_1 \circ \mathcal{I}_1. \tag{14}$$

This averaged map is CPTP. Write its affine Bloch action as $\boldsymbol{r} \mapsto \boldsymbol{t} + T\boldsymbol{r}$, equivalently $C(\sigma_j) = \sum_i T_{ij}\sigma_i$ for Pauli matrices σ_j.

Every positive trace-preserving map contracts the trace norm on Hermitian inputs. Indeed, the adjoint is positive and unital and therefore preserves the operator order interval $[-I, I]$ used in the variational characterisation of the Hermitian trace norm. Triangle inequality in Eq. (14) then yields

$$2\|T\boldsymbol{e}_j\|_2 = \|C(\sigma_j)\|_1 \le \sum_{b=0}^{1} \|\mathcal{I}_b(\sigma_j)\|_1. \tag{15}$$

The equality on the left uses trace preservation: $C(\sigma_j)$ is a traceless Hermitian qubit operator.

For the instrument (8), direct evaluation gives

$$\begin{aligned} \mathcal{I}_0(X) &= psX, & \mathcal{I}_1(X) &= qsX, \\ \mathcal{I}_0(Y) &= psY, & \mathcal{I}_1(Y) &= qsY, \\ \mathcal{I}_0(Z) &= \mathrm{diag}(p, \gamma - p), & \mathcal{I}_1(Z) &= \mathrm{diag}(q - \gamma, -q). \end{aligned} \tag{16}$$

Because $q \le \gamma \le p$, the two Z-image trace norms sum to $2p$. Equation (15) therefore bounds the three Bloch columns by

$$\|T\boldsymbol{e}_x\|_2 \le s, \qquad \|T\boldsymbol{e}_y\|_2 \le s, \qquad \|T\boldsymbol{e}_z\|_2 \le p, \tag{17}$$

and hence

$$\|T\|_F^2 \le 2(1-\gamma) + p^2 = \frac{1199}{625}. \tag{18}$$

For a two-use current-bit policy, summing both records factorises exactly:

$$\Phi_2^M = C_2 \circ C_1, \qquad C_k = \sum_{b=0}^{1} R_{k,b} \circ \mathcal{I}_b. \tag{19}$$

No unitality or unitary restriction is used here. For trace-preserving qubit maps with Bloch blocks T_1, T_2, the normalised entanglement fidelity of the composition is

$$F_e(C_2 \circ C_1) = \frac{1 + \mathrm{Tr}(T_2 T_1)}{4}. \tag{20}$$

The affine translation does not contribute to the superoperator trace underlying Eq. (20). Frobenius Cauchy–Schwarz and Eq. (18) now give

$$\mathrm{Tr}(T_2 T_1) \le \|T_2\|_F \|T_1\|_F \le \frac{1199}{625}, \tag{21}$$

which proves Eq. (9). Combining the global upper bound with Eq. (13) yields

$$\Delta_2 \ge \frac{1831}{2500} - \frac{456}{625} = \frac{7}{2500} > 0. \tag{22}$$

At one use, a record history contains only the current bit, so the two policy classes coincide and $\Delta_1 = 0$. This completes the proof of Theorem 1.

5 Interpretation and limitations

The theorem shows that an environment bit available for immediate recovery need not remain adequate for the next recovery: retaining the earlier coarse bit can improve the two-use optimum. The mechanism is not Hilbert-space size alone. Coarse graining turns each disclosed outcome into a two-Kraus operation, removing the resolved record on which the existing qubit no-advantage argument relies [1, 2]. The separation remains valid even when every current-bit recovery is an arbitrary qubit CPTP map.

Several conclusions are deliberately excluded. The proof does not determine either exact optimum F_2^M or F_2^B; its $7/2500$ is a certified lower bound on their difference, not the exact gap. It does not optimise over instruments or establish a smallest positive gap on the continuous rank-four domain. It supplies no robustness radius for perturbations or laboratory errors and makes no claim about an $n > 2$ optimum. Finally, the literature comparison is bounded rather than an exhaustive priority search; the relation to broader partially observed quantum control should be reviewed by the author before submission.

Reproducibility

The exact witness, Pauli images, rank certificate, trace-norm bounds, hidden-path calculation, and an independent Liouville evaluation are implemented in the accompanying Python script. The calculations use rational and symbolic algebra for the logical certificate; floating-point output is only a replay. The accompanying reproducibility file records the environment and invocation.

Statements and declarations

Funding. No funding information was supplied for this draft; the author must confirm or replace this statement before submission.

Competing interests. No competing-interest information was supplied for this draft; the author must provide the final declaration before submission.

Data and code availability. No empirical data were used. The exact input record and verification code are included with the local manuscript package.

AI use declaration. OpenAI `gpt-5.6-sol` was used on 13–14 September 2026 to assist problem formulation, bounded literature discovery, derivation, code generation, exact and numerical cross-checks, and manuscript drafting. All outputs remain subject to human verification; the exact calculations were also replayed in independent representations against the stated equations and primary-source records. AI is not an author and cannot assume accountability. This local draft requires independent review, correction, and approval by the named human author before any submission.

References

[1] M. Gregoratti and R. F. Werner. On quantum error-correction by classical feedback in discrete time. *Journal of Mathematical Physics*, 45(7):2600–2612, 2004. doi: 10.1063/1.1758320.

[2] Milajiguli Rexiti, Samad Khabbazi Oskouei, and Stefano Mancini. On the effectiveness of Bayesian discrete feedback for quantum information reclaiming. *Quantum Information Processing*, 23(7):258, 2024. doi: 10.1007/s11128-024-04467-x.

[3] H. M. Wiseman, Stefano Mancini, and Jin Wang. Bayesian feedback versus markovian feedback in a two-level atom. *Physical Review A*, 66:013807, 2002. doi: 10.1103/PhysRevA.66.013807.